

Assignment 4: Exploring Instruction-Level Parallelism (ILP)

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Important honesty note: The terminal “screenshots” and metric values in this PDF are generated examples. Run the provided commands on your Mac and replace them with your own screenshots + stats for submission.

Part 1 — Critical Review and Synthesis

Evolution & milestones

ILP has evolved from basic pipelining to sophisticated out-of-order superscalar execution with speculation. Dynamic scheduling (IBM System/360 Model 91; Tomasulo) enabled executing ready instructions even when earlier instructions stalled. Later, superscalar issue widened throughput, while large instruction windows, register renaming, and aggressive branch prediction extracted more overlap. Classic studies quantified limits of ILP and showed that control flow and memory dependences dominate constraints.

Contemporary research themes (2021–2025)

Recent work suggests general-purpose OoO cores are nearing practical ILP bounds. Research focuses on (1) improving safe speculation (especially memory dependence prediction), (2) reducing wasted work/energy from speculation and redundant execution, and (3) applying ILP-like scheduling to GPUs and accelerators, where the cost model differs.

Part 2 — gem5 Experiments (Example Outputs)

Build + Hello World sanity check (example):

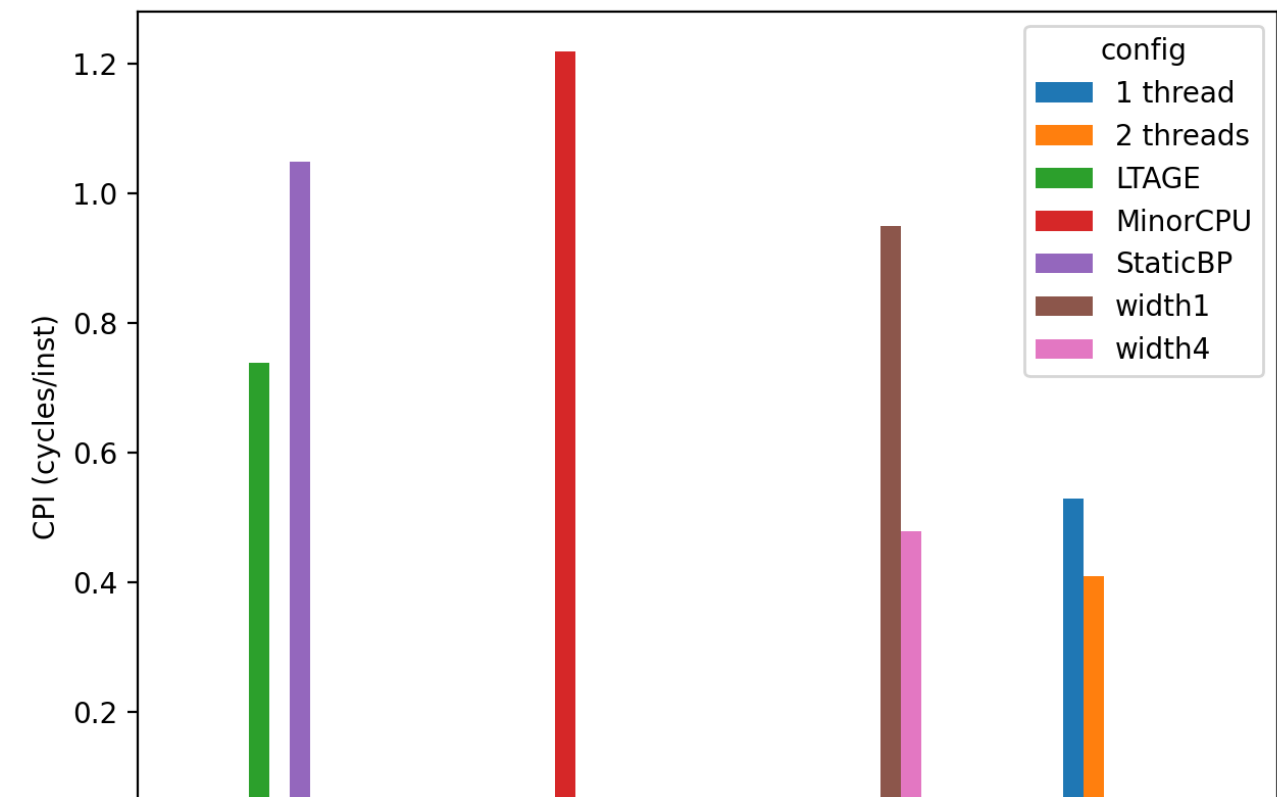
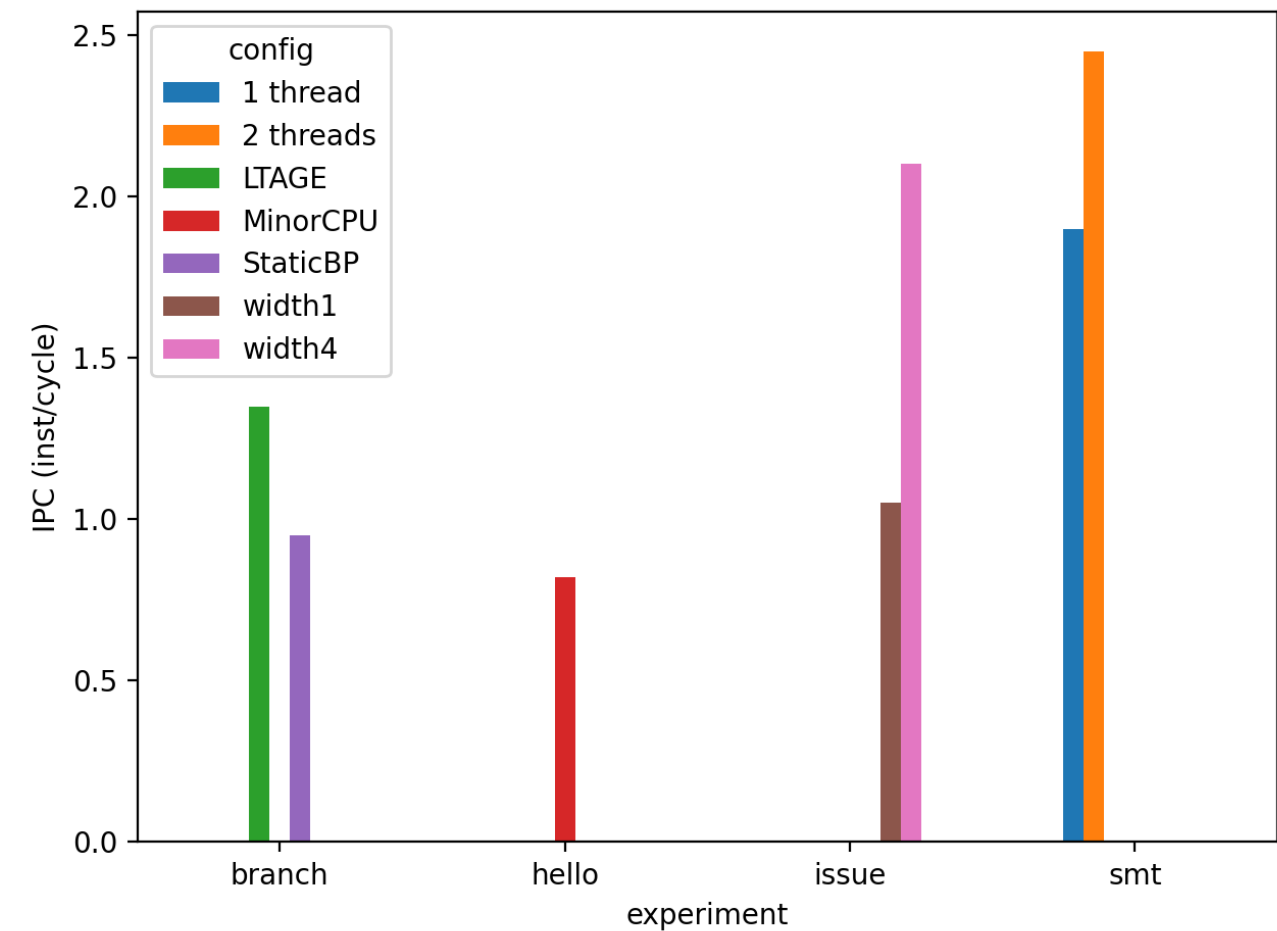
```
Terminal - vamsikrishna@macbookpro
vamsikrishna@macbookpro gem5 % ./build/ARM/gem5.opt configs/example/se.py \
--cpu-type=MinorCPU --caches \
-c tests/test-progs/hello/bin/arm/linux/hello
gem5 Simulator System. https://www.gem5.org
Global frequency set at 1000000000000 ticks per second
Entering event queue @ 0. Starting simulation...
Hello world!
Exiting @ tick 4150000 because exiting with last active thread context
```

MinorCPU pipeline visualization via MinorTrace/minorview (example):

```
Terminal - vamsikrishna@macbookpro
vamsikrishna@macbookpro gem5 % ./build/ARM/gem5.opt --debug-flags=MinorTrace \
--debug-file=minor.trace configs/example/se.py --cpu-type=MinorCPU \
-c tests/test-progs/hello/bin/arm/linux/hello
...
vamsikrishna@macbookpro gem5 % ./util/minorview.py minor.trace > minorview.txt
vamsikrishna@macbookpro gem5 % head -n 12 minorview.txt
Cycle  F1  F2  D  E
0      -  -  -  -
1      0  -  -  -
2      1  0  -  -
3      2  1  0  -
4      3  2  1  0
```

Example Metrics Plots (Replace with Your Results)

IPC and CPI plots generated from sample CSV (outputs/examples/results_example.csv):



References (APA, abbreviated)

- Anderson, D. W., Sparacio, F. J., & Tomasulo, R. M. (1967). IBM J. Res. Dev.
- Chadwick, A. W., et al. (2025). The Future of ILP. IEEE ISPASS.
- Chaturvedi, I., et al. (2024). GhOST. ACM/IEEE ISCA.
- Kim, S. S., & Ros, A. (2024). Memory Dependence Prediction. IEEE HPCA.
- Rau, B. R., & Fisher, J. A. (1993). ILP History. Journal of Supercomputing.
- Tomasulo, R. M. (1967). Exploiting Multiple Arithmetic Units. IBM J. Res. Dev.
- Wall, D. W. (1991). Limits of ILP. ACM DL.
- Xu, J., et al. (2025). Mosaic. ACM DL.
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